Computational Accelerator Physics

Pavel Snopok

PHYS 540

Welcome!

Let's start with three quotes:

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– D. Adams, The Hitchhiker's Guide to the Galaxy

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– V. P. Denissov, my third-year elective class instructor

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Studying is fun!

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– Martin Berz, my Ph.D. advisor

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- Final project: 30-minute presentation (25+5) and written report on the topic of your choice (25% of the final grade)

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- Any questions before we move on?

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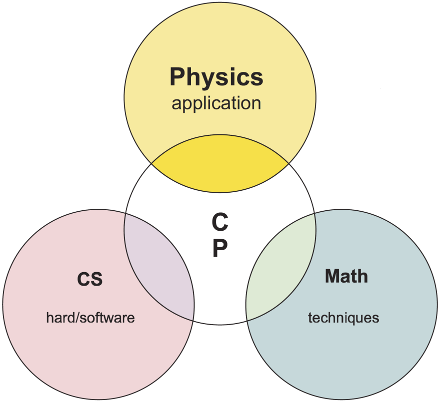
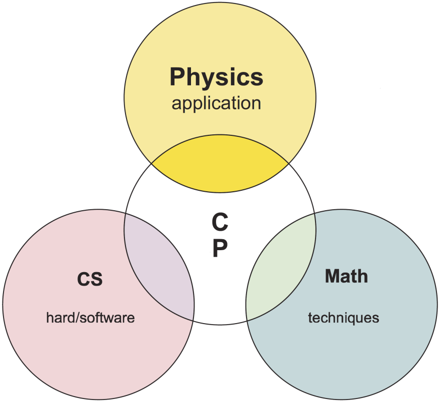
- Nowadays, I am heavily involved with neutrino experiments at Fermilab (NOvA/DUNE)

Computational Physics





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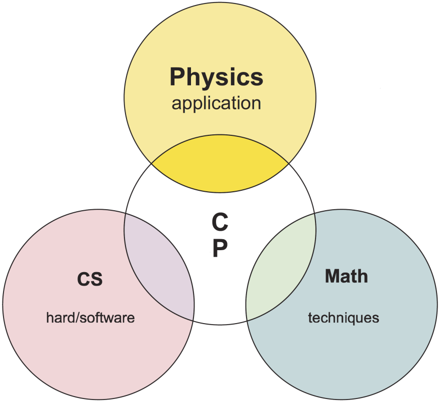

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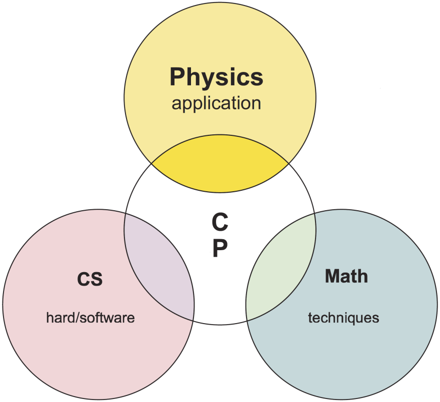

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- ...and we are good to go!

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- Accelerator vs beam physics, do you feel the difference?

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- Open any textbook on accelerator physics...

History of accelerators - electrostatic

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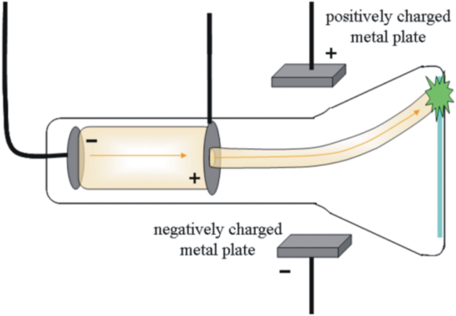

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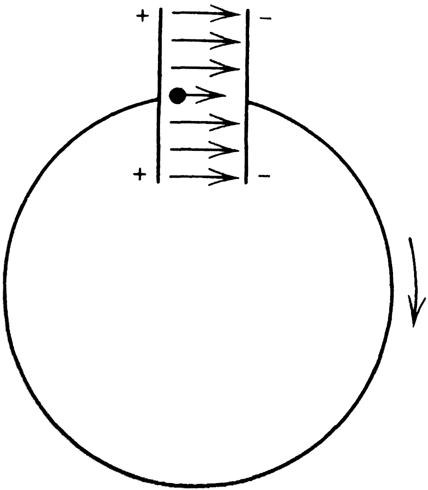

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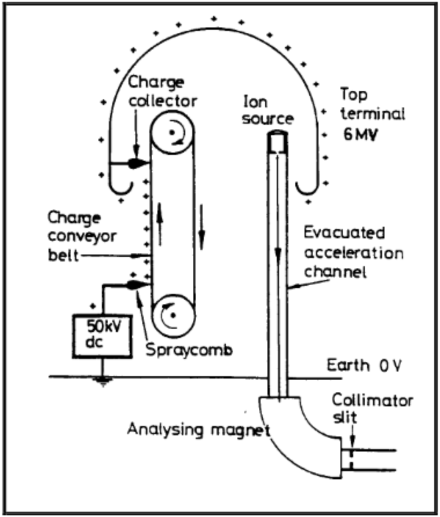

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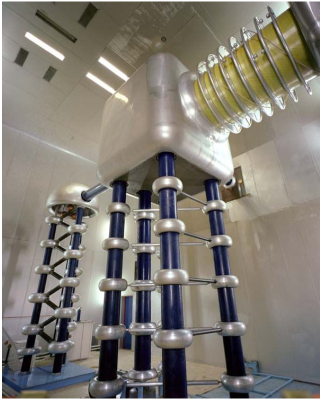

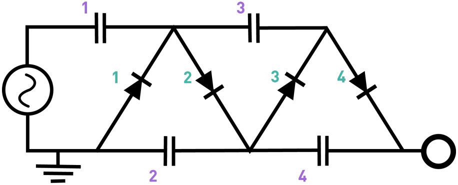

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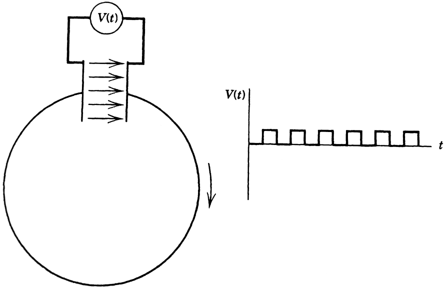

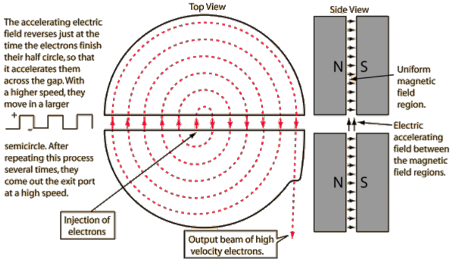

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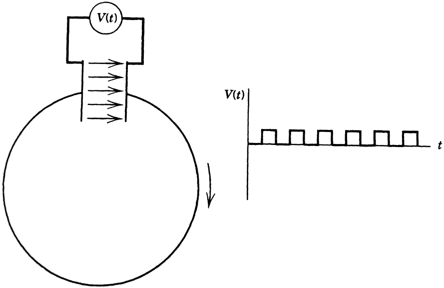

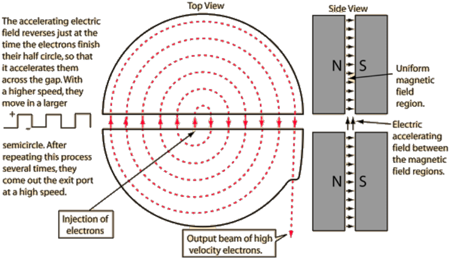

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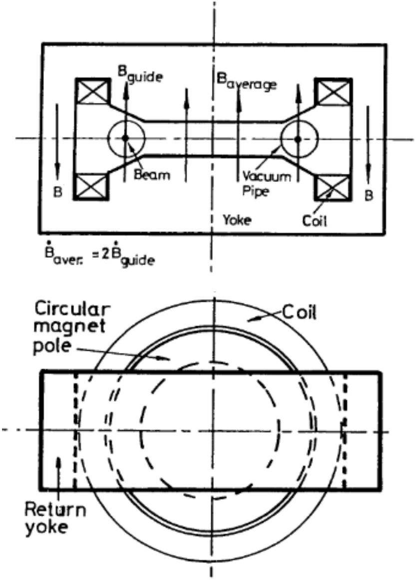

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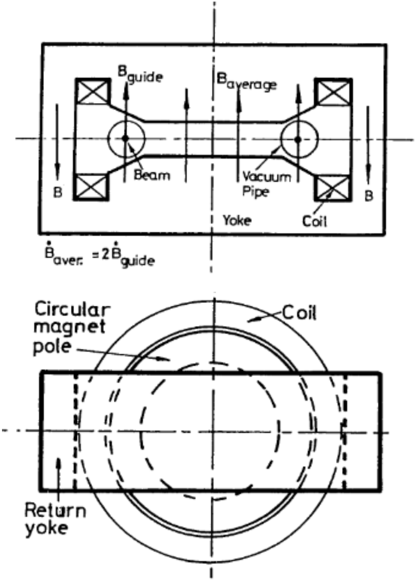

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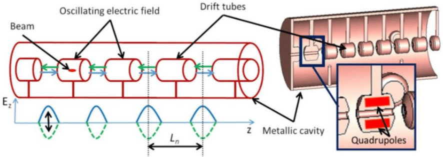

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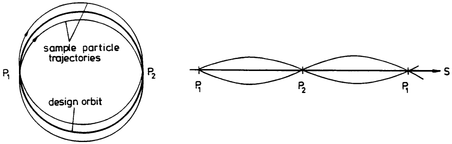
- Synchrotron ⇒ strong (or "alternating gradient") focusing!

weak focusing:



strong focusing:

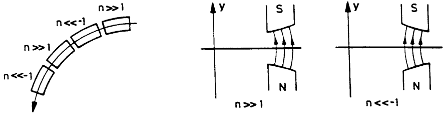


Figure 5: Alternating-gradient focusing

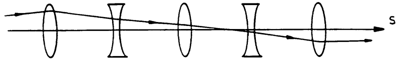


Figure 6: An alternating series of focusing and defocusing lenses leads to overall focusing if the distances between the lenses are not too large.

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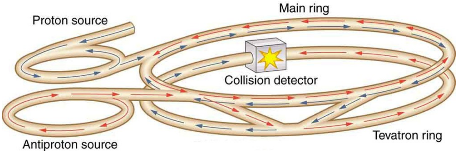

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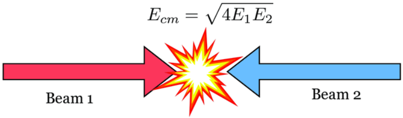
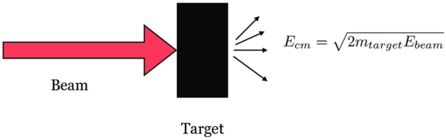
- Collider vs fixed target experiment – what's the big deal?





Stanford Linear Accelerator Center (SLAC NAL)

- The SLAC research program centers on experimental and theoretical research in elementary particle physics using electron beams
- The 2.0 mile long underground accelerator is the longest linear accelerator in the world, and is claimed to be “the world’s straightest object”
- 50 GeV RF electron-positron collider

Radio Frequency Quadrupole

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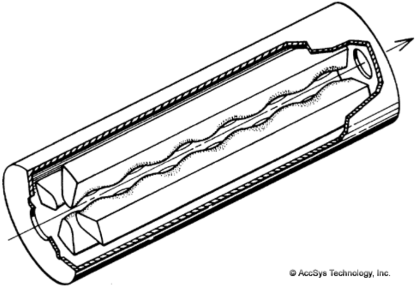
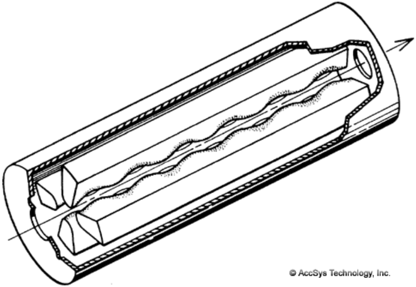
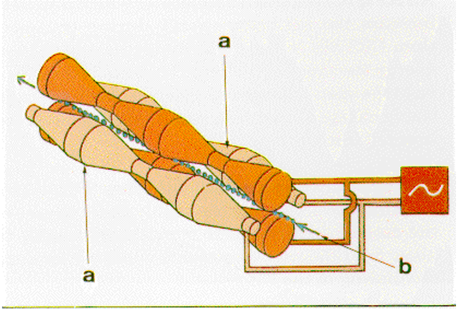


- RFQ general structure

Radio Frequency Quadrupole

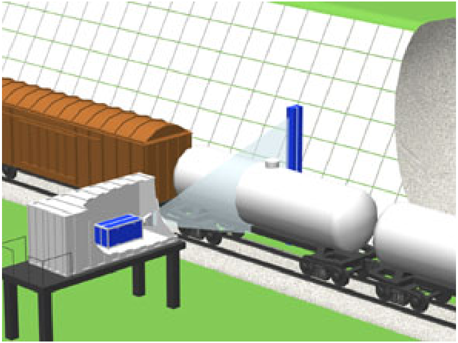




- RFQ general structure
- RFQ detailed view

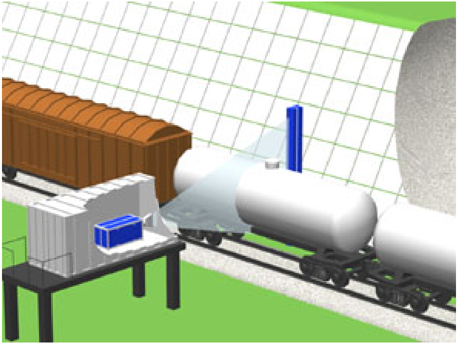
One possible application

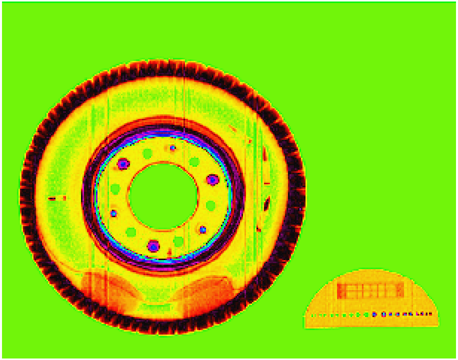
One possible application



- Scanning train

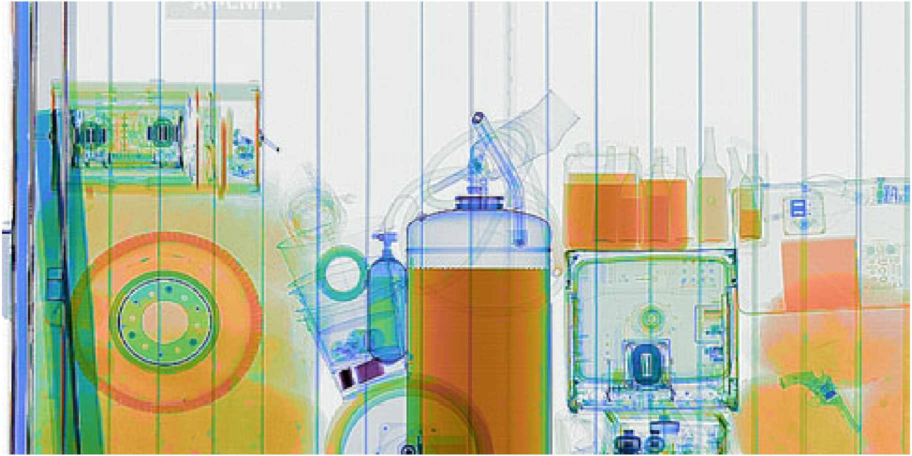
One possible application





- Scanning train
- Contraband found

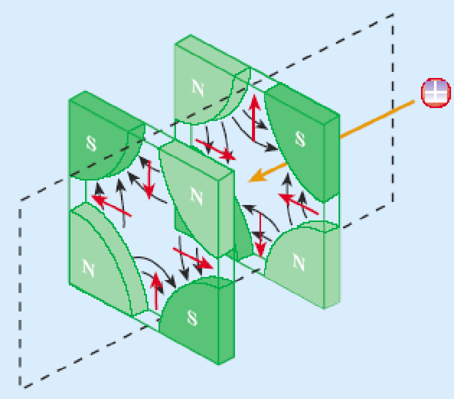
Dual energy scanning



Orange – organics, blue – metals, magenta – heavy metals, green – organic and non-organic overlap

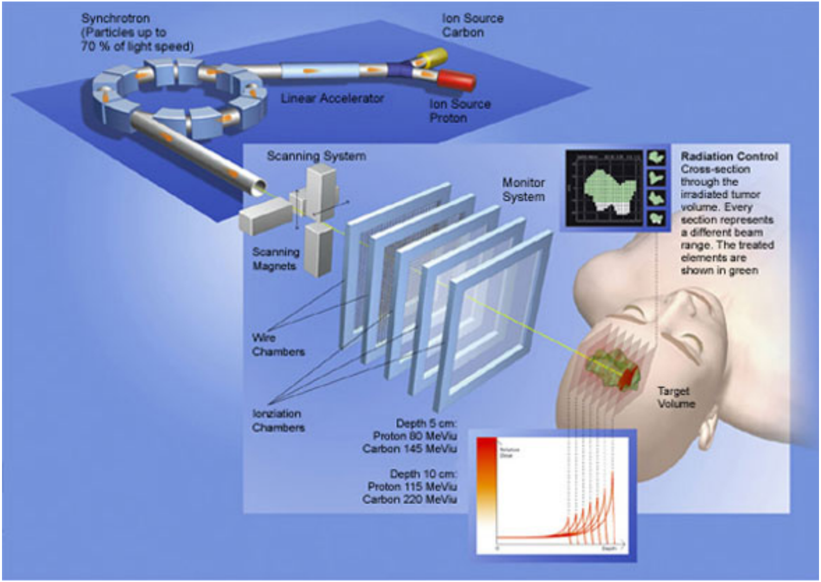
Radiation therapy with Cyclotrons

- Cyclotrons can be used to treat cancer
- Ion beams penetrate the body and kill tumors by radiation damage, while minimizing damage to healthy tissue along their path
- Cyclotron beams can be used to bombard other atoms to produce short-lived positron-emitting isotopes suitable for PET imaging

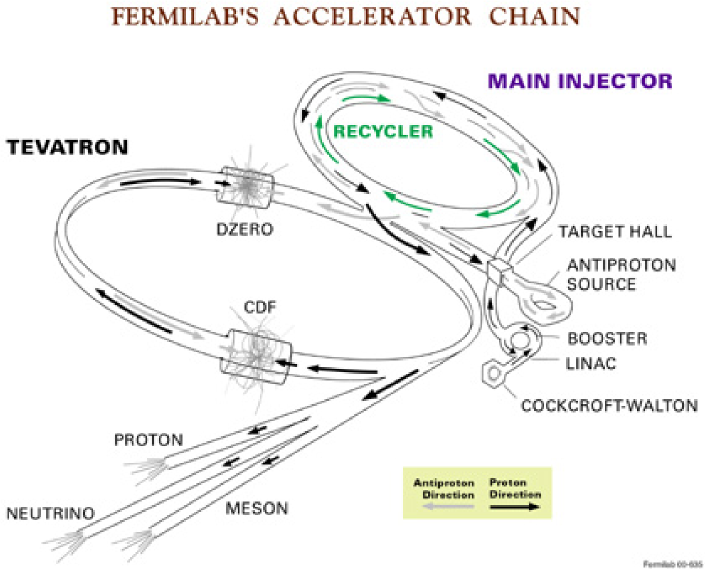


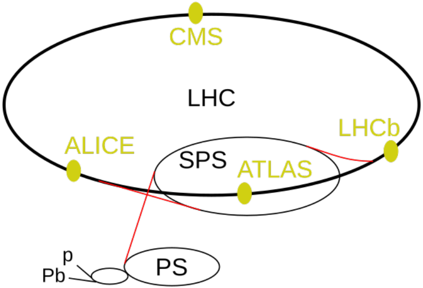
- Synchrotrons – way to get past the limitations of the cyclotrons (non-relativistic velocities, closer orbits, getting out of sync)
- Strong focusing principle
- BNL (1957) – Courant and Snyder
- Increased beam intensity
- Reduced the overall construction cost of a more powerful accelerator

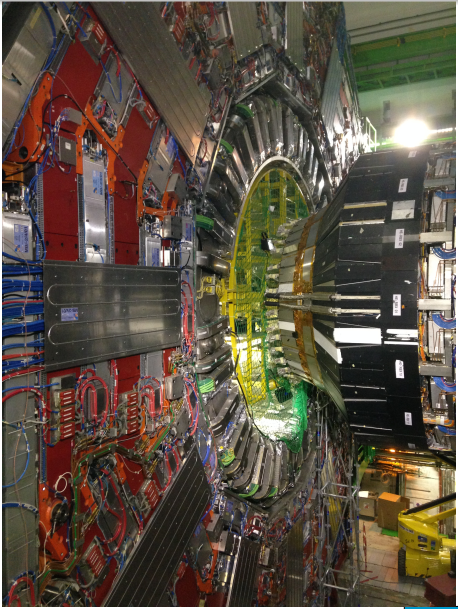
Radiation therapy with synchrotrons



Tevatron Accelerator Complex (before 2012)







- p, Pb: linear accelerators
- Booster (the small unmarked circle), Proton Synchrotron (PS), Super Proton Synchrotron (SPS)
- LHC: 27-km-long tunnel
- 4 large experiments marked with yellow dots (CMS: figure on the right)

Two types of accelerators

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- Linear (Need to know as much as possible about the nonlinearities, particles pass through the accelerator once)

Two types of accelerators

- Linear (Need to know as much as possible about the nonlinearities, particles pass through the accelerator once)
- Circular (Need to know as much as possible about the long-term stability (return later), particles pass through the accelerator many times)

Steering and Focusing Charged Particles

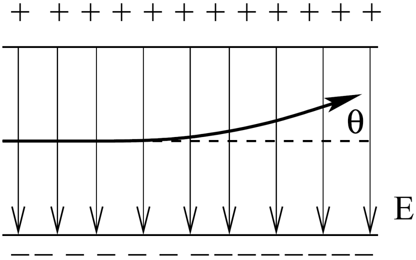
Steering and Focusing Charged Particles I

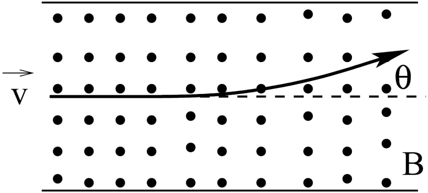
- In order to form the beam of particles and keep it tight for the duration of the acceleration cycle, we need three basic types of elements:
  - Guiding elements: keep the particles on a well-defined *reference orbit*
  - Focusing elements: keep the particles within the vacuum chamber close to the reference orbit
  - Accelerating elements: give the particles a boost to get to the design energy

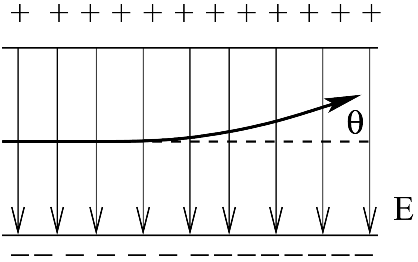
- Assume we have a charged particle, and we need to choose whether we use $\boldsymbol{E}$ or $\boldsymbol{B}$ to steer off its straight track:

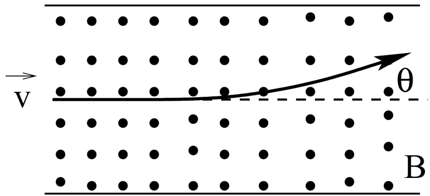
$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

- What would be more efficient/appropriate?





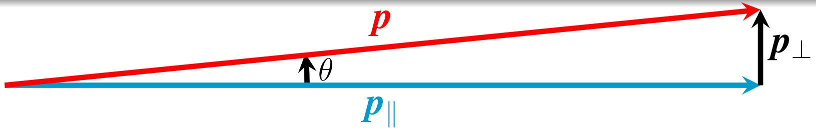




Algorithm:

- Consider paraxial approximation (appropriate in most cases)
- Derive expression for small angle change as the particle steps through the $\boldsymbol{E}$ or $\boldsymbol{B}$ field
- Integrate to find the overall change in angle per unit length
- Substitute E or B into the resulting expression
- Compare the outcomes and make conclusions about the suitability

Bending particle trajectory II



- $\boldsymbol{p}_\parallel \gg \boldsymbol{p}_\perp$ ($\boldsymbol{p}_\perp$ is exaggerated in the picture), so θ is small:

$$\tan(\theta) \approx \sin(\theta) \approx \theta \approx \frac{p_\perp}{p_\parallel} \approx \frac{p_\perp}{p}$$

- Denote small change in transverse momentum dp, then

$$d\theta = \frac{dp}{p} = \frac{1}{p}\frac{dp}{dt}\frac{dt}{dz}dz = \frac{1}{p}F\frac{1}{v}dz$$

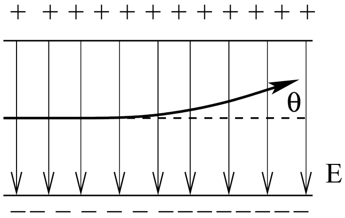
- Thus, the total angle change

$$\theta = \frac{1}{pv}\int_0^L F dz$$

Bending particle trajectory III

$$\theta = \frac{1}{pv}\int_0^L F dz$$

Let's substitute some numbers: assume we deal with electrons with total energy 10 GeV, and we want to bend it through 0.01 rad over 1 m. In this case relativistic $\beta = 0.995$ and energy over charge is 10 GeV/e = 10^{10} V.



- Let's assume the particle overall direction of motion is $\hat{z}$, and we want to bend the particle in the $\hat{\boldsymbol{x}}$ direction. Then $\boldsymbol{E}$ should be in the $-\hat{\boldsymbol{x}}$ direction, and $\boldsymbol{B}$ – in the $\hat{\boldsymbol{y}}$ direction (verify that!)

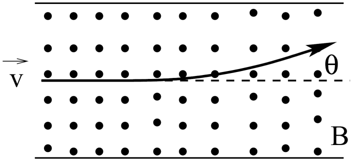
- If we use $\boldsymbol{E}$: $F_E\hat{\boldsymbol{x}} = -eE_x(-\hat{\boldsymbol{x}}) = eE_x\hat{\boldsymbol{x}}$ and $\theta_{\text{total}} = \frac{1}{pv}\int_0^L eE_x dz$

$$E_x L = \int_0^L E_x dz = \frac{\theta p v}{e} = \frac{\theta(\gamma m c^2)\beta^2}{e} = (10\,\text{mrad})(10^{10}\,\text{V})(0.99) = 99\,\text{MV}$$

Bending particle trajectory IV

$$\theta = \frac{1}{pv}\int_0^L F dz$$

Let's substitute some numbers: assume we deal with electrons with total energy 10 GeV, and we want to bend it through 0.01 rad over 1 m. In this case relativistic $\beta = 0.995$ and energy over charge is 10 GeV/e = 10^{10} V.



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- Same for $\boldsymbol{B}$: $F_B\hat{\boldsymbol{x}} = -ev\hat{\boldsymbol{z}} \times B_y\hat{\boldsymbol{y}} = evB_y\hat{\boldsymbol{x}}$ and $\theta_{\text{total}} = \frac{1}{pv}\int_0^L evB_y dz$

$$B_yL = \int_0^L B_y dz = \frac{\theta p}{e} = \frac{\theta(\gamma mc^2)\beta}{ec} = \frac{(10\,\text{mrad})(10^{10}\,\text{V})(0.995)}{3 \times 10^8\,\text{m/s}} = 0.33\,\text{T}$$

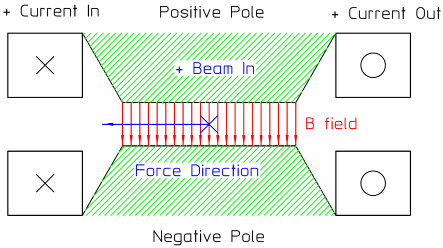
- Summary: $E_x = 99$ MV/m is tough while $B_y = 0.33$ T/m is reasonable.

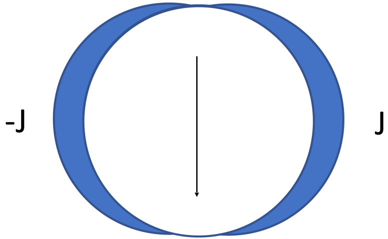
Magnets

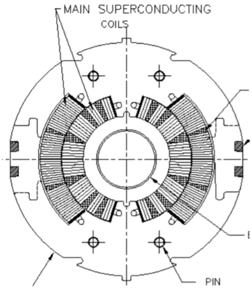
- So we agreed: we bend and focus using the magnetic field
- But there is a catch: magnetic forces do no work ($\boldsymbol{F} \perp \boldsymbol{v}$) $\Rightarrow$ use electric field for acceleration
- Different magnet shapes:
  - Dipole
  - Quadrupole
  - Sextupole
  - N-pole
  - Combined function
  - Solenoid
  - ...
- Different materials:
  - Iron-dominated: shaped by the iron
  - Superconducting: shaped by coil placement
  - Superferric: combination of both

Bending particles: dipole magnet

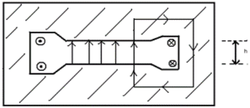
- Two poles, constant field, steers particle beam

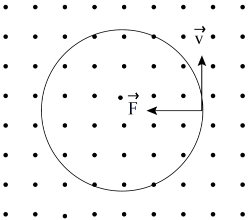






Dipole magnet field and motion

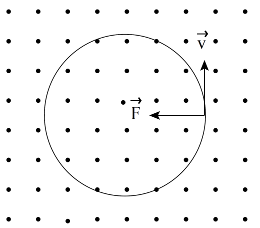




- The field in the gap near the axis (where the beam is) is nearly uniform.
  - Field calculation = try it at home (hint: in the next few slides I'll show how to do this for a quadrupole magnet). Your answer should be

$\boxed{B_0 = \dfrac{\mu_0 n I}{h},}$ where n – # of turns in the coil, I – coil current, h – size of the gap

- Uniform magnetic field $\Rightarrow$ circular motion
  - ...and it is intrinsically unstable
  - That's where combined-function magnet helps, but let's look at the quad first



- Let's derive an important relation from the equation of motion in the uniform magnetic field
- Two forces are in balance: the Lorentz force $F_L = evB$ ($v \perp B$) and the centrifugal force $F = \gamma m_0 v^2/\rho$, where ρ is the radius of curvature
- Thus,

$$F = \gamma m_0 \frac{v^2}{\rho} = qvB$$

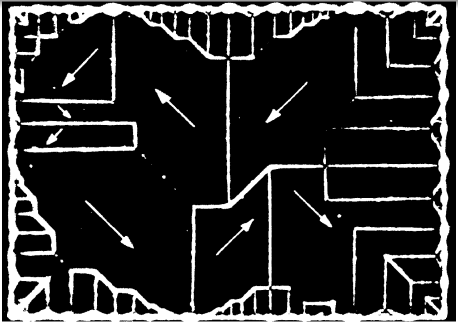
-

$$\boxed{\frac{p}{q} = B\rho} \text{ or } \boxed{B\rho\,[\text{T}\cdot\text{m}] = \frac{p\,[\text{GeV}/c]}{0.2998}}$$

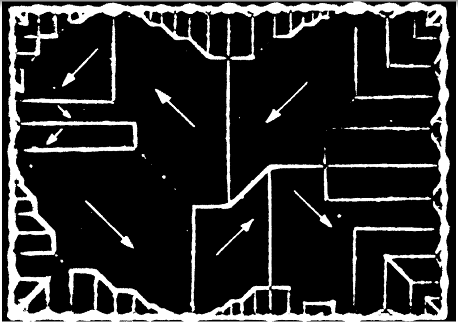
- The ratio of p to q describes the "stiffness" of a beam, it can be considered as a measure of how much angular deflection results when a particle travels through a given magnetic field
  - For a specific magnetic field, the greater the momentum of a particle, the less it will be bent as it travels through that field
  - The greater the charge of a particle, the more it will be bent as it travels through a given magnetic field

Brief recap from PHYS 413: Magnets

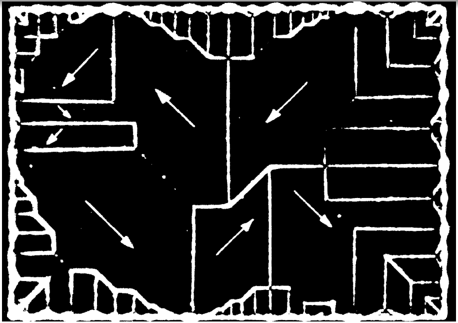
Ferromagnetism



Ferromagnetism



- Ferromagnets: keep magnetization in the absense of the external field



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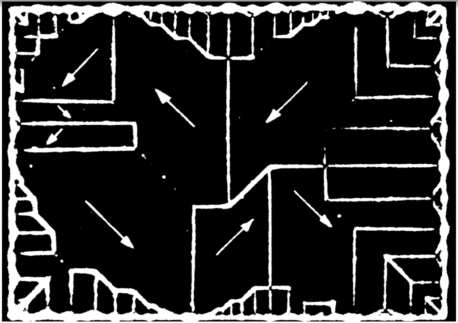
Ferromagnetism



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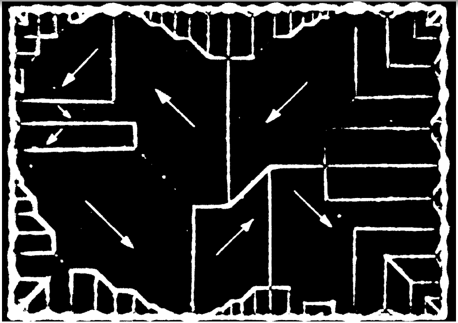
- Dipoles tend to line up in the absence of magnetic field and form domains.

Ferromagnetism



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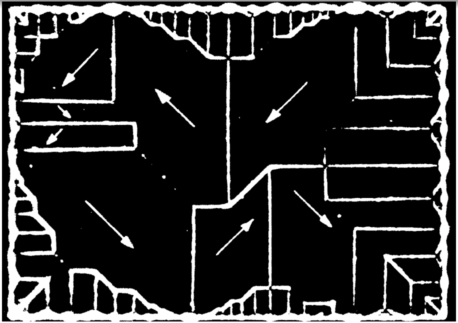
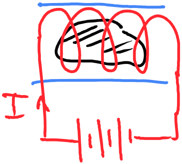
- In the presence of a strong external field, domains aligned with the field grow and stay that way.

Ferromagnetism



- Ferromagnets: keep magnetization in the absense of the external field $\Rightarrow$ permanent magnets.
- Dipoles tend to line up in the absence of magnetic field and form domains.
- In the presence of a strong external field, domains aligned with the field grow and stay that way.
- If the field is strong enough, one domain wins, and the material (iron) is saturated.

Consider a solenoid with a chunk of iron inside:



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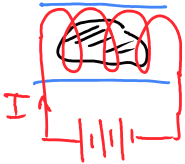

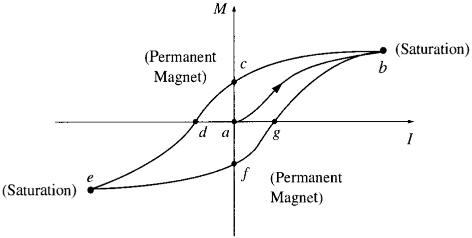

- Start at zero current, $M = 0$.

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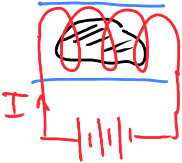

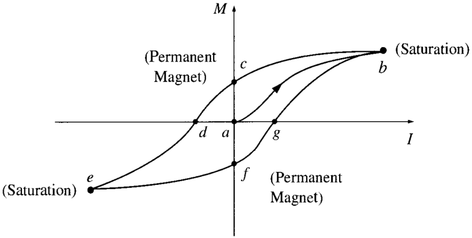

- (a) Start at zero current, $M = 0$.
- (b) Increase current until saturation (max M) is achieved.

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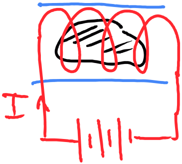

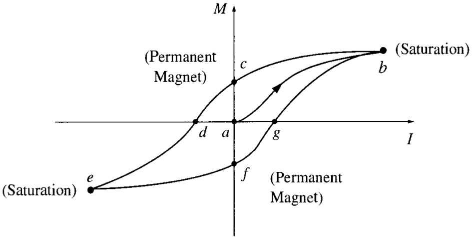

- a Start at zero current, $\boldsymbol{M} = 0$.
- b Increase current until saturation (max $\boldsymbol{M}$) is achieved.
- c Reduce current to zero $\Rightarrow$ some dipoles reorient, but not all $\Rightarrow$ residual magnetization $\boldsymbol{M}$.

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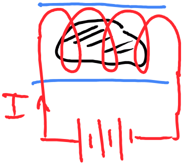

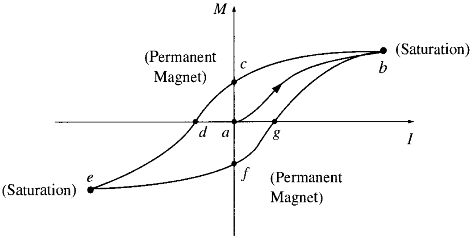

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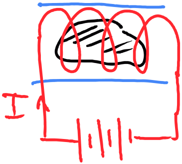
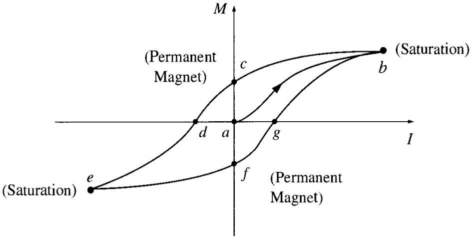
- d. Turn the current around, $I < 0 \Rightarrow \boldsymbol{M} = 0$ at some point.

Hysteresis loop

Consider a solenoid with a chunk of iron inside:





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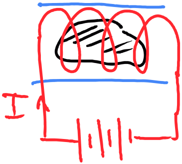

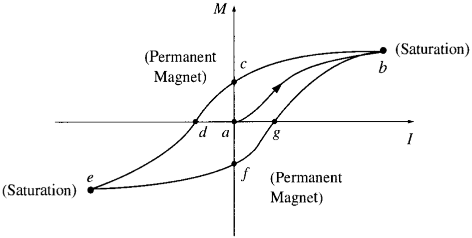

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Reverse the process to get back to point b.

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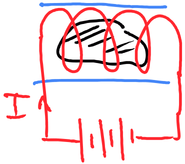

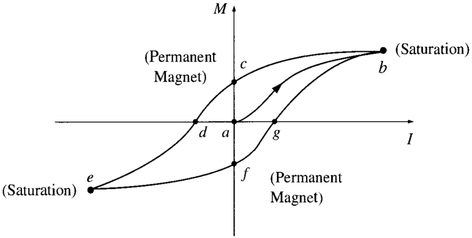

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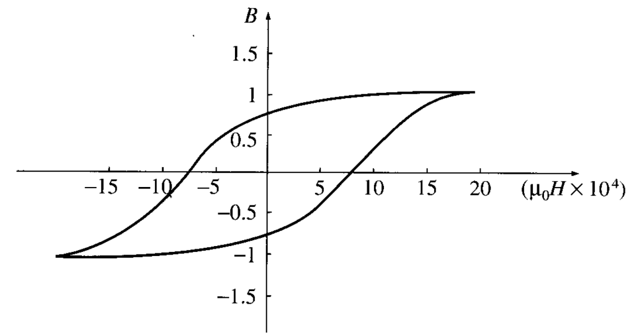
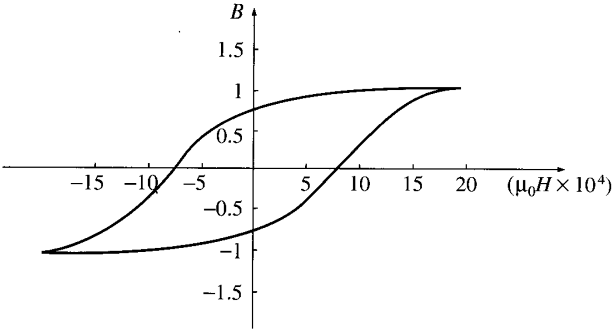
Note. $\boldsymbol{M}$ depends on the history: $I = 0$ at a, c, f, yet $\boldsymbol{M}$ is different.

Different perspective



Different perspective



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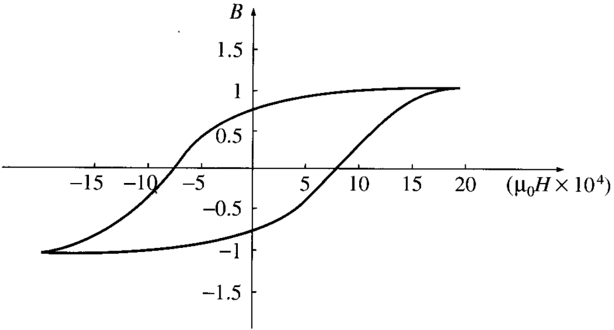

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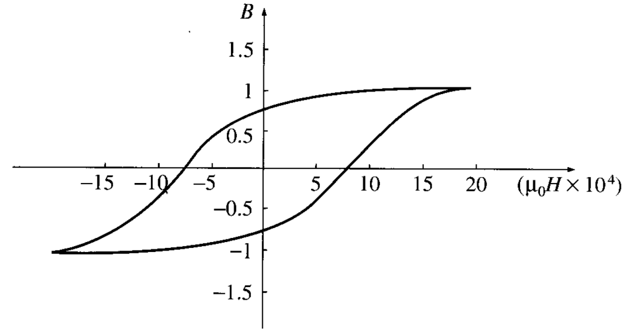

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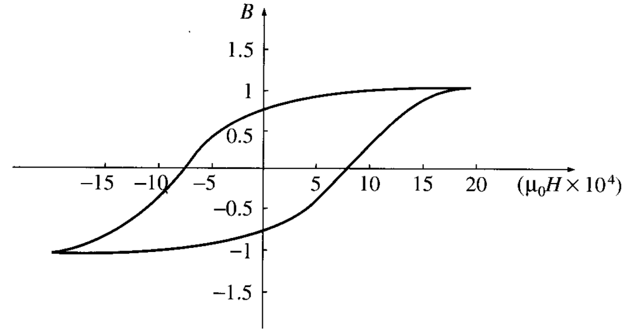

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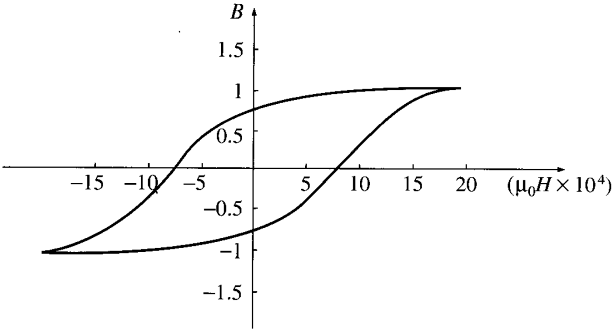

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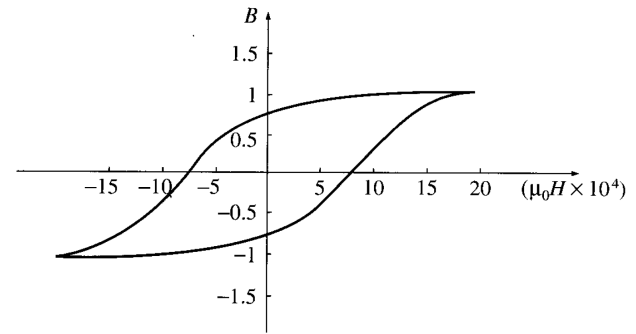

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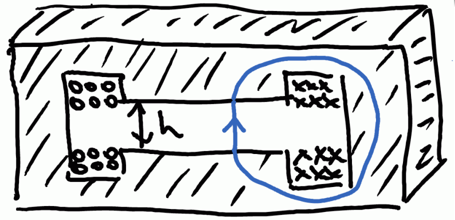
- $\mu/\mu_0 \sim 10^4$, $\Rightarrow \mu_0 H \ll B$, where $\mu_0 H$ is the field with no iron and B is the actual field with the iron core.

Example: dipole magnet



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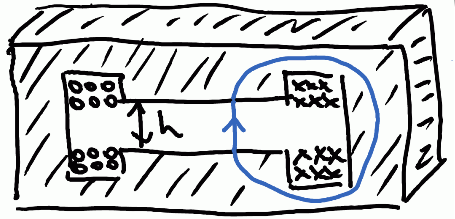

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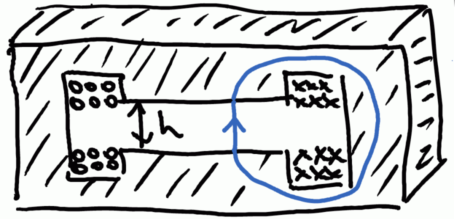

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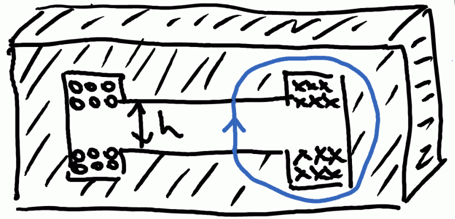

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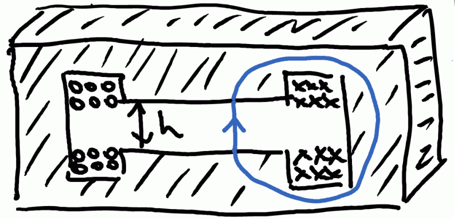

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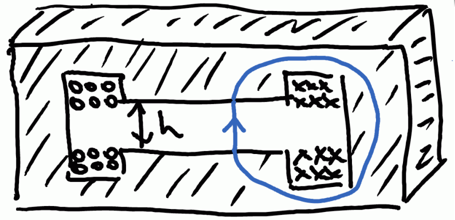

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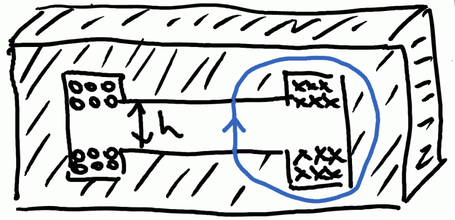

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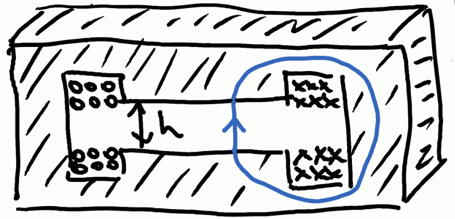

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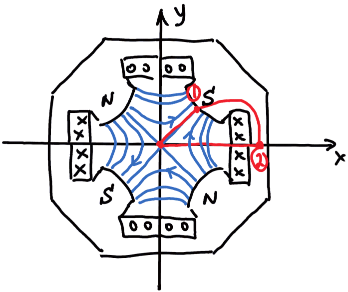

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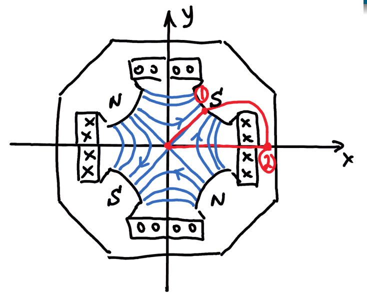

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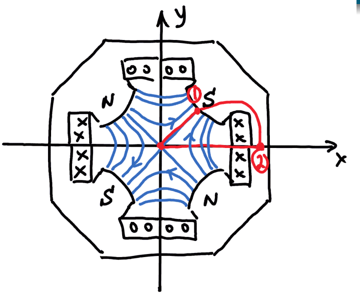

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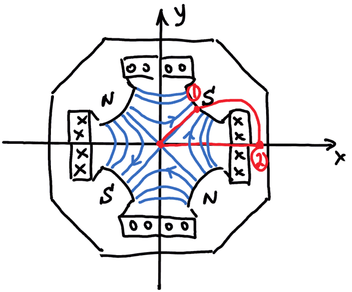

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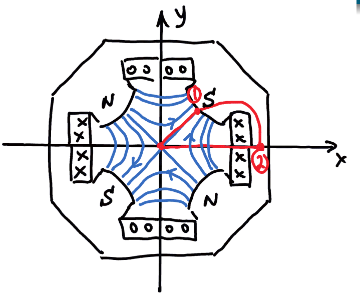

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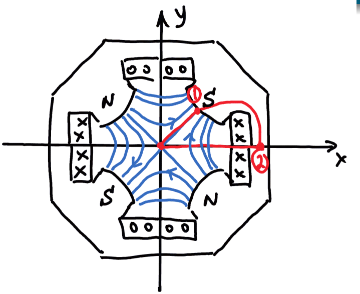

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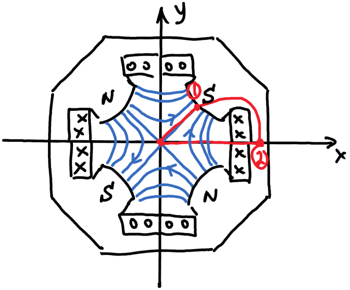

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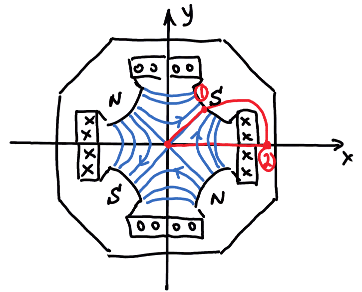

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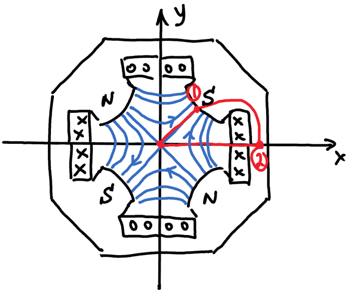

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Back to the main topic: accelerator codes

Accelerator Codes

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Bmad

Let's take a quick look at the structure of the Bmad manual/tutorials...